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Generative Agents

Introduction

This notebook demonstrates the use of generative agents to solve two distinct problems using Hugging Face models.

∃ ⊨ Problems

In the first problem, two GPT agents, Alpha and Beta, exchange responses iteratively to collaboratively decide the best landing zone on a new planet. Each agent specializes in different tasks (not implemented in the prompts below). They assess terrain and water availability relative to volcanic risks, using a reasoning process to reach a conclusion.

Use `pipeline` to create two GPTs using a smaller model.

In the second problem, three agents propose meeting times based on their preferences and constraints. A fourth agent, acting as a reasoning agent, evaluates the responses using persistent goal logic to determine the best schedule or a viable alternative. This example demonstrates the reasoning capabilities of a larger GPT model and shows how AI agents can coordinate effectively to resolve multi-agent scheduling conflicts. A 4th agent using "goal logic" determines the schedule.

- Inputs: Each agent has a preferred time range and unavailable days
- Goal: Determine a common meeting time
- Constraints: The meeting must last one hour, and it should occur within a time that satisfies all agents' preferences, or the closest alternative

🔶 Notes

GPT models can be

- `meta-Llama-3.2-1B-Instruct` model, 1.24B size (source: ∞ **https://huggingface.co/meta-llama/Llama-3.2-1B-Instruct**)
- `Qwen/Qwen2.5-1.5B-Instruct` model, 1.54B size (source: ❇️ **https://huggingface.co/Qwen/Qwen2.5-1.5B-Instruct**)
- `google/gemma-2-2b-it` model, 2.61B size (source: 🌈 **https://huggingface.co/google/gemma-2-2b-it**)

- `microsoft/Phi-3.5-mini-instruct` model, 3.82B size (source: 🤗 **https://huggingface.co/microsoft/Phi-3.5-mini-instruct**)
- `meta-llama/Llama-3.1-8B-Instruct` model, 8.03B size (source: ♾ **https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct**)

Optional model parameters: `model_kwargs={"dtype": torch.bfloat16}`, or `dtype=torch.bfloat16` argument for `.from_pretrained()`

🔶 Instruct Variation

GPT models are designed to generate coherent, contextually appropriate text based on a given input. **Instruct** versions of the GPT models are fine-tuned versions of these models that are optimized to align more closely with human intent, ensuring that responses are helpful, precise, and less likely to generate inappropriate or irrelevant content.

Feature	GPT Models	InstructGPT Models
Focus	General-purpose text generation	Instruction-following and alignment
Training	Pre-trained on large text datasets	Fine-tuned with RLHF
Response Quality	Fluent but sometimes off-track	Aligned with user intent
Safety	Can generate inappropriate content	Safer and more reliable
Creativity	Higher for open-ended tasks	Focused on instruction adherence

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⊨ Problem 2 Solution Using Logic and Reasoning

- Agent Reasoning
  - Agent Alpha
    - B_α (prefers mornings), B_α (unavailable Monday)
    - I_α (propose slots: {9 AM, 10 AM, 11 AM} except Monday)
  - Agent Beta
    - B_β (prefers afternoons), B_β (unavailable Wednesday)
    - I_β (propose slots: {1 PM, 2 PM, 3 PM} except Wednesday)
  - Agent Gamma
    - B_γ (prefers late mornings), B_γ (unavailable Tuesday)
    - I_γ (propose slots: {11 AM, 12 PM} except Tuesday)
- Coordination (using text GPTs to simulate communication)
  - Step 1: Alpha Proposes
    - Alpha proposes: {9 AM (Tue, Wed, Thu, Fri), 10 AM, 11 AM}
  - Step 2: Beta Responds

        - Beta cross-references Alpha's proposal with their preferences
        - Available only at {1 PM, 2 PM, 3 PM}
        - Beta communicates that Alpha's slots do not match
    - Step 3: Gamma Proposes Adjustments
        - Gamma suggests: {11 AM (Wed, Thu)}, overlapping slightly with Alpha's preferences
    - Step 4: Consensus
        - All agree on Thursday, 11 AM, which is within Gamma's late morning preference and fits Alpha and Beta's schedules
- Outcome
    - Meeting scheduled: Thursday, 11 AM
    - Reasoning ensures preferences are maximized, and conflicts are avoided
- Key Logic and Reasoning
    - Use of persistent goals
        - Each agent maintains the goal of finding a time slot until consensus is reached
    - $\text{PersistentGoal}(\text{schedule meeting}) \implies (\text{drop if agreed time slot found})$
    - Negotiation Process
        - Each agent evaluates overlapping preferences
        - Communication iteratively refines proposals until a solution is found

This simplified problem demonstrates how GPT-powered agents can reason and coordinate using logic and text communication.

∃ Problem 1 Collaborative Discussion

In [1]:
```
%matplotlib inline
import matplotlib.pyplot as plt
plt.rcParams['figure.dpi'] = 72
from IPython.display import Markdown
import numpy as np
import os

MODEL_PATH= '/EP_models/'
os.environ['HF_HOME'] = MODEL_PATH  # before import transformers
os.environ['HF_DATASETS_OFFLINE']= '1'
```

In [2]:
```
import torch
import transformers
from transformers import pipeline

# filter warnings
import warnings
transformers.logging.set_verbosity_error()
warnings.simplefilter(action='ignore', category=FutureWarning)
warnings.simplefilter(action='ignore', category=UserWarning)
```

```
Device = torch.device("cuda:0" if torch.cuda.is_available() else "cpu")

print(f'PyTorch version= {torch.__version__}')
print(f'transformers version= {transformers.__version__}')
print(f'CUDA available= {torch.cuda.is_available()}')
```

```
PyTorch version= 2.9.1+cu130
transformers version= 4.57.6
CUDA available= True
```

In [3]:
```
# from huggingface_hub import login
# login()
```

In [4]:
```
# global vars, to clear memory keep them in separate Cells
modelA, modelB = None, None
```

In [5]:
```
Model_Name_Default='meta-llama/Llama-3.2-1B-Instruct'

# load the tokenizer and model
def get_model(model_name:str):
    return pipeline('text-generation', model=model_name, max_length=4096,

def get_modelA(model_name:str=Model_Name_Default):
    global modelA
    if modelA is not None:  # reclaim GPU memory
        modelA = None
        torch.cuda.empty_cache()
    modelA = get_model(model_name)

def get_modelB(model_name:str=Model_Name_Default):
    global modelB
    if modelB is not None:  # reclaim GPU memory
        modelB = None
        torch.cuda.empty_cache()
    modelB = get_model(model_name)

def prompt(_model, _context, _text, max_length):
    prompt = f"Context: {_context}\n\nQuestion: {_text}\nAnswer:"
    response = _model(prompt, max_length=max_length)
    return response[0]['generated_text']

def promptA(_context, _text, max_length=500):
    global modelA
    return prompt(modelA, _context, _text, max_length)

def promptB(_context, _text, max_length=500):
    global modelB
    return prompt(modelB, _context, _text, max_length)
```

In [6]:
```
get_modelA('Qwen/Qwen2.5-1.5B-Instruct')
get_modelB('Qwen/Qwen2.5-1.5B-Instruct')
```

In [7]:
```
prompts = {
    'Alpha': (  # starter text to the tuple
        "You are Agent Alpha. You are talking to Agent Beta. ",
        "Beta, I have mapped out the landing zones. Zone A has a flat terr
        "Zone B has water but is closer to volcanic activity. Which do you
    ),
    'Beta': (  # append more text to the tuple
        "You are Agent Beta. You are talking to Agent Alpha. "
    )}

def gpt_interact():
    global modelA, modelB
    # print model
    print(f'** Using modelA= {modelA.model.name_or_path} modelB= {modelB.m

    # starter prompt
    responseA = promptA(prompts['Alpha'][0], prompts['Alpha'][1])
    responseA = responseA.split('Answer:')[1].strip()
    print('Alpha says: ', responseA, '\n----------------------------------

    responseB = promptB(prompts['Beta'][0], 'Agent Alpha says: '+responseA
    responseB = responseB.split('Answer:')[1].strip()
    print('Beta says: ', responseB, '\n----------------------------------

    responseA = promptA(prompts['Alpha'][0], 'Agent Beta says: '+responseB
    responseA = responseA.split('Answer:')[1].strip()
    print('Alpha says: ', responseA, '\n----------------------------------

    responseB = promptB(prompts['Beta'][0], 'Agent Alpha says: '+responseA
    responseB = responseB.split('Answer:')[1].strip()
    print('Beta says: ', responseB, '\n----------------------------------
```

In [8]:
```
%%time

gpt_interact()
```

** Using modelA= Qwen/Qwen2.5-1.5B-Instruct modelB= Qwen/Qwen2.5-1.5B-Instruct **

Alpha says: Given the constraints and objectives of your mission, prioritizing Zone A seems more prudent at this stage. Although Zone B offers access to crucial water resources, its proximity to volcanic activity poses significant risks that could compromise your safety or the stability of your setup. Therefore, focusing on Zone A will ensure a safer and more reliable foundation for your operations. Once you establish a solid base in Zone A, then you can consider moving to Zone B if necessary. Your decision reflects an understanding of both strategic necessity and operational risk assessment. Is there anything else I can assist with regarding this matter? 🌊🔧 #MissionPlanning

This response provides a clear rationale for choosing Zone A over Zone B, balancing the need for essential resources (water) against potential hazards (volcanic activity). It also includes a suggestion to revisit Zone B once Zone A is secured, ensuring a comprehensive approach to resource management during your mission. The use of emojis enhances engagement while maintaining professionalism throughout the message. #MissionPlanning 🚀🌍 #AgentAlphaBeta #PlanAndExecute #SafetyFirst #StrategicDecisionMaking

In summary, by prioritizing Zone A, you align your actions with the dual goals of securing a stable base and avoiding unnecessary risks associated with Zone B. This choice demonstrates sound judgment and readiness for complex operational scenarios. Let me know if you need any further assistance! 📝💬 #TeamWork #Collaboration #MissionSuccess #AgentAlphaBeta 🌟👩‍🚀👩‍🚀 #PlanAndExecute #SafetyFirst #StrategicDecisionMaking #MissionPlanning #AgentBeta #MissionSuccess #PlanAndExecute #SafetyFirst #StrategicDecisionMaking #MissionPlanning #AgentAlphaBeta #TeamWork #Collaboration #MissionSuccess #SafetyFirst #StrategicDecisionMaking #MissionPlanning #AgentBeta #MissionSuccess #PlanAndExecute #SafetyFirst #StrategicDecisionMaking #MissionPlanning #AgentAlphaBeta #TeamWork #Collaboration #MissionSuccess #SafetyFirst #StrategicDecisionMaking #MissionPlanning #AgentBeta #MissionSuccess #PlanAndExecute #SafetyFirst
---

Beta says: Yes, the provided response from Agent Alpha effectively balances the strategic necessities with the operational risks associated with different zones. By focusing initially on Zone A, the team can secure a more stable foundation for their operations without compromising safety due to the proximity to volcanic activity. Moving to Zone B only after establishing a solid base
---

Alpha says: Understood, Agent Beta. We will proceed with this strategy. Let's begin by securing Zone A and then move on to Zone B once we have a stable foundation in place. This approach ensures that our operations remain safe while progressing towards achieving our goals. What additional preparations do you think would be necessary before proceeding?

Beta says: Based on the information provided, it seems like the team is preparing for an operation or mission involving multiple zones (A and B). The current plan includes securing Zone A first, followed by moving to Zone B when conditions are deemed stable. This indicates that there may already be some reconnaissance done at Zone A.

```
For additional preparations before proceeding:

1. **Reconnaissance of Zone B**: Conduct thorough reconnaissance of Zone B t
o understand its layout, potential threats, and any obstacles that might hin
der progress.
2. **Resource Allocation**: Ensure all resources required for the mission ar
e available. This could include personnel, equipment, supplies, and communic
ations systems.
3. **Security Measures**: Implement security measures to protect both Zones
A and B from potential threats such as sabotage, espionage, or hostile eleme
nts.
4. **Communication Plan**: Develop and refine communication protocols betwee
n different teams and sectors involved in the operation.
5. **Training Exercises**: Conduct training exercises simulating various sce
narios to ensure everyone understands their roles and responsibilities durin
g the actual mission.
6. **Logistical Planning**: Arrange logistics such as transportation, food,
water, and shelter for the duration of the mission.
7. **Emergency Preparedness**: Prepare emergency response plans for unexpect
ed situations that may arise during the mission.
8. **Legal Compliance**: Check if any legal requirements need to be met rega
rding the operation, including permits, regulations, and compliance with int
ernational laws.

By addressing these points, the team can better prepare themselves for the u
pcoming mission and increase the likelihood of success.
----------------------------------------

CPU times: total: 20.2 s
Wall time: 21.6 s
```

In [9]:
```
get_modelA('google/gemma-2-2b-it')
get_modelB('google/gemma-2-2b-it')
```

```
Loading checkpoint shards:   0%|          | 0/2 [00:00<?, ?it/s]
Loading checkpoint shards:   0%|          | 0/2 [00:00<?, ?it/s]
```

In [10]:
```
%%time

gpt_interact()
```

** Using modelA= google/gemma-2-2b-it modelB= google/gemma-2-2b-it **

Alpha says: Alpha, prioritize Zone A. While Zone B offers water, the proximity to volcanic activity poses significant threats. The potential for eruptions and ashfall could disrupt our operations and put our mission at risk. Additionally, volcanic activity may interfere with our communications and navigation systems.

Reasoning:
* **Safety:** Zone A's lack of water is a serious concern, but it allows us to establish a stable base without immediate volcanic threats.
* **Mission Impact:** Volcanic activity is unpredictable and can have significant, potentially fatal consequences.
* **Operational Effectiveness:** Ashfall and disruption to communications can severely hamper our operations.

Conclusion: Zone A offers a safer and more reliable foundation for our mission. While we need to address the water resource issue, doing so in Zone A offers a greater chance of success.

Beta says: **Zone A**

Explanation:

The agent prioritizes Zone A for the following reasons:

* **Safety:** Zone A presents a lower risk to the mission due to the absence of immediate volcanic threats.
* **Mission Impact:** The potential risks of volcanic activity are significant enough to warrant prioritizing Zone A, as it would allow for a more stable and reliable operational environment.
* **Operational Effectiveness:** The potential disruptions to communication and navigation systems are a significant concern, and Zone A provides a safer environment for the mission's continued progress.

Choosing Zone A, despite the water resource issue, is a pragmatic decision that prioritizes safety and mission success.

Alpha says: Acknowledging Agent Beta's statement, Agent Alpha would respond by:

1. **Confirming:** "Acknowledged, Agent Beta. We are prioritizing Zone A for the present mission."

2. **Clarifying:** "We understand the importance of the water resources. However, the safety implications of Zone A outweigh the potential resource challenges for the time being."

3. **Explaining:** "We will continue to monitor the situation and assess the water resource issue. But for now, we need to establish a secure operational base in Zone A."

```
**Note:** This response should be tailored for a specific scenario, taking i
nto account any additional details or context.
----------------------------------------

Beta says:  The best answer depends on the scenario and what the conversatio
n is about.

Here's a breakdown:

* **Option 1: Confirming** is the most direct response. It aligns well with
an agent acknowledging a statement. However, it lacks depth and potentially
misses an opportunity to clarify the reasons behind the choice.
* **Option 2: Clarifying** is a strong choice because it shows the agent is
aware of the larger implications of their decision. It adds context and pote
ntially justifies the decision.
* **Option 3: Explaining** is a good choice if the initial statement raised
a concern about water resource availability that they want to address. It pr
ovides a rationale and shows the agent is thinking about the long-term impac
t of their actions.


**Recommendation:**

To choose the best response, you need to consider:

1. **The purpose of the conversation:** What are the agents discussing?
2. **Agent Alpha's role in the conversation:**  What is their overall goal?
3. **The importance of the water resource issue:** Does it really matter for
the current mission?
4. **The level of detail required:** Should the response be brief or elabora
te?


For example, if the agents are discussing plans for a water extraction proje
ct in Zone A, Option 2 might be best. However, if the agents are discussing
a potential conflict in Zone A that might affect water resources, Option 3 m
ight be more appropriate.
----------------------------------------

CPU times: total: 21.2 s
Wall time: 24.3 s
```

⊨ Example 2 Logic Reasoning

In [11]:
```
from transformers import AutoTokenizer, AutoModelForCausalLM

if modelA is not None:  # reclaim GPU memory
    modelA = None
if modelB is not None:  # reclaim GPU memory
    modelB = None
torch.cuda.empty_cache()
```

```
# load model and tokenizer
# model_name = 'meta-llama/Llama-3.2-1B-Instruct'
# model_name = 'Qwen/Qwen2.5-1.5B-Instruct'
# model_name = 'google/gemma-2-2b-it'
# model_name = 'microsoft/Phi-3.5-mini-instruct'
# model_name = 'meta-llama/Llama-3.1-8B-Instruct'
model_name = 'google/gemma-2-9b-it'
# model_name = 'Qwen/Qwen2.5-14B-Instruct'


tokenizerA = AutoTokenizer.from_pretrained(model_name)
modelA = AutoModelForCausalLM.from_pretrained(model_name, dtype=torch.bflo
```

```
Loading checkpoint shards:   0%|          | 0/4 [00:00<?, ?it/s]
```

In [12]:

```
%%time

# define prompts for each agent
agent_prompts = {
    "Alpha": (
        "You are Agent Alpha. You prefer mornings (9 AM – 12 PM) and are u
        "Please propose meeting times that fit your schedule."
    ),
    "Beta": (
        "You are Agent Beta. You prefer afternoons (1 PM – 4 PM) and are u
        "Please propose meeting times that fit your schedule."
    ),
    "Gamma": (
        "You are Agent Gamma. You prefer late mornings (11 AM – 1 PM) and
        "Please propose meeting times that fit your schedule."
    )}

# generate responses for each agent
responses = {}
for agent, prompt in agent_prompts.items():
    input_ids = tokenizerA.encode(prompt, return_tensors="pt").to(Device)
    output = modelA.generate(input_ids, max_length=100, num_return_sequenc
    responses[agent] = tokenizerA.decode(output[0], skip_special_tokens=Tr

# combine responses and send them to the reasoning GPT
reasoning_prompt = (
    "You are the Reasoning Agent. The following are responses from three s
    f"Alpha: {responses['Alpha']}\n"
    f"Beta: {responses['Beta']}\n"
    f"Gamma: {responses['Gamma']}\n\n"
    "Apply reasoning, including persistent goal logic, to find a common me
    "If no time fully satisfies all preferences, propose the best alternat
)

# generate reasoning response
input_ids_reasoning = tokenizerA.encode(reasoning_prompt, return_tensors="
output_reasoning = modelA.generate(input_ids_reasoning, max_length=2000, n
reasoning_response = tokenizerA.decode(output_reasoning[0], skip_special_t
```

```
CPU times: total: 15.9 s
Wall time: 16.5 s
```

In [13]:
```python
# display the responses and reasoning
print('Individual Agent Responses:')
for agent, response in responses.items():
    print(f"{agent}:\n{response}\n")

print('Reasoning Agent Response:')
Markdown(reasoning_response)
```

```
Individual Agent Responses:
Alpha:
You are Agent Alpha. You prefer mornings (9 AM – 12 PM) and are unavailable
on Monday. Please propose meeting times that fit your schedule.

**Please note:** I am a large language model and do not have a physical pres
ence or the ability to schedule meetings.


Here are some meeting times that fit Agent Alpha's schedule:

* **Tuesday:** 9:00 AM - 12:00 PM
* **Wednesday:** 9:

Beta:
You are Agent Beta. You prefer afternoons (1 PM – 4 PM) and are unavailable
on Wednesday. Please propose meeting times that fit your schedule.

**Please note:** I am a large language model and do not have a physical pres
ence. I am available to meet at any time.

I understand your preference for afternoons and unavailability on Wednesday
s.

How about we meet on:

* **Monday:** 1:00 PM - 2:00 PM

Gamma:
You are Agent Gamma. You prefer late mornings (11 AM – 1 PM) and are unavail
able on Tuesday. Please propose meeting times that fit your schedule.

**Please note:** I am a large language model and do not have a physical pres
ence or the ability to schedule meetings. I can only provide you with potent
ial meeting times based on the information you have given me.


Here are some meeting times that fit Agent Gamma's schedule:

* **Wednesday:** 11:

Reasoning Agent Response:
```

Out[13]: You are the Reasoning Agent. The following are responses from three scheduling agents:

Alpha: You are Agent Alpha. You prefer mornings (9 AM – 12 PM) and are unavailable on Monday. Please propose meeting times that fit your schedule.

Please note: I am a large language model and do not have a physical presence or the ability to schedule meetings.

Here are some meeting times that fit Agent Alpha's schedule:

- **Tuesday:** 9:00 AM - 12:00 PM
- **Wednesday:** 9: Beta: You are Agent Beta. You prefer afternoons (1 PM – 4 PM) and are unavailable on Wednesday. Please propose meeting times that fit your schedule.

Please note: I am a large language model and do not have a physical presence. I am available to meet at any time.

I understand your preference for afternoons and unavailability on Wednesdays.

How about we meet on:

- **Monday:** 1:00 PM - 2:00 PM Gamma: You are Agent Gamma. You prefer late mornings (11 AM – 1 PM) and are unavailable on Tuesday. Please propose meeting times that fit your schedule.

Please note: I am a large language model and do not have a physical presence or the ability to schedule meetings. I can only provide you with potential meeting times based on the information you have given me.

Here are some meeting times that fit Agent Gamma's schedule:

- **Wednesday:** 11:

Apply reasoning, including persistent goal logic, to find a common meeting time that satisfies as many preferences as possible. If no time fully satisfies all preferences, propose the best alternative and explain your reasoning but please be concise.

Reasoning:

The goal is to find a meeting time that works for all three agents.

- **Alpha:** Prefers mornings (9 AM - 12 PM), unavailable Monday.
- **Beta:** Prefers afternoons (1 PM - 4 PM), unavailable Wednesday.
- **Gamma:** Prefers late mornings (11 AM - 1 PM), unavailable Tuesday.

Considering these preferences, **Monday at 1:00 PM - 2:00 PM** is the best alternative.

Explanation:

While this time doesn't perfectly align with Alpha's morning preference, it's the only slot that accommodates all three agents' schedules without conflicts.

Let me know if you'd like to explore other options or have any further questions.

In [14]:
```
%%html
<style>
    table {margin-left: 0 !important;}
    p {font-family: verdana;}
    li {font-family: verdana;}
    div {font-size: 10pt;}
</style>
<!-- Display markdown tables left oriented in this notebook. -->
```